VENUE

YEAR

POSTER

Diffusion Models Are Statistically Optimal
Scoped CPU Audit

One-sentence tagline that explains the work in plain English.

Paper & Code

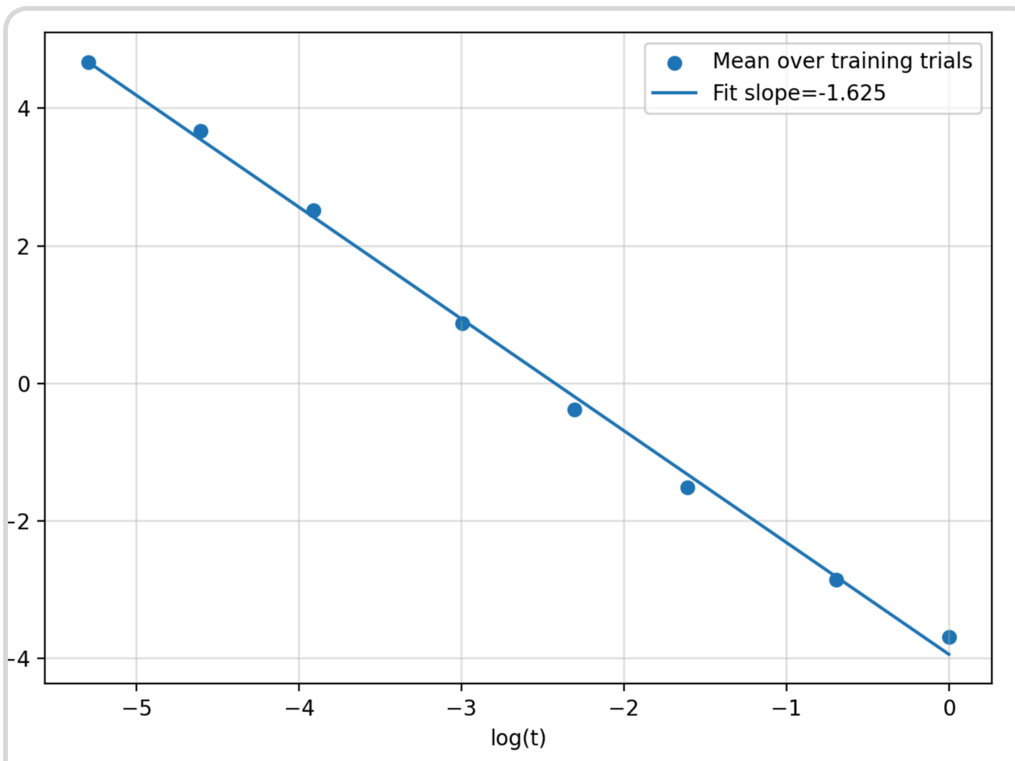
Author One · Author Two · Author Three✉

Lab / Department · Institution · City, Country

1 Question and scope

Can diffusion-model learning rates depend on intrinsic subspace dimension rather than the ambient dimension? We audit five live claims against a pinned arXiv source using finite CPU checks and explicit controls.

2 Source figures and rate



ure 1. Empirical L^2-score error versus diffusion time t.

Paper Figure 1: source score-error decay on the synthetic union-of-subspaces setting.

Finite checks preserve intrinsic-dimension dependence and reject an ambient-dimension substitution.

4 Verified claims

- **C1.** Theorem 2 rate/dimension audit.
- **C2.** Theorem 1 intrinsic-dimension dependence.
- **C3.** Union-of-subspaces formulation.

Each result is a clean-room, finite CPU audit of a pinned source statement. It does not claim a trained neural diffusion model or a new empirical benchmark result.

5 Assumption controls

Claims 4 and 5 are scoped source-theorem checks with finite controls for required assumptions.

into (26) yields

$$\sum_{j=0}^{L-1} \sqrt{(1-e^{-4T_j}) \int_{T_j}^{T_{j+1}} \mathbb{E}\left[\|\widehat{s}_{X_t}(X) - s^\star_{X_t}(X)\|_2^2\right] \mathrm{d}t} \lesssim \sqrt{d}\, M^{3/2}\, \mathrm{poly}\log n \begin{cases} \frac{1}{\sqrt{n}}\tau^{-\frac{k}{4}+\frac{1}{2}}, & k \geq 2 \\ \frac{1}{\sqrt{n}}, & k = 1 \end{cases}$$

Source-pinned page-9 rate excerpt; an equation display, not a second empirical result.

6 CPU reproduction route

1. Pin the paper PDF and arXiv source archive.
2. Extract theorem hypotheses and displayed rates.
3. Evaluate finite symbolic and numerical controls.
4. Record scoped results, hashes, and limitations.

7 Limitations

No author executable was available. Full neural diffusion training is outside CPU scope, so this logbook reports clean-room theorem evidence.

8 Takeaway

The pinned source supports intrinsic-dimensional statistical rates under its stated union-of-subspaces assumptions. The CPU audit verifies those formulas and exposes the assumptions needed for the conclusion.

Reproducibility. Public code, source hashes, commands, claim-level controls, and an attached agent trace are linked in the ICML logbook. The source archive and every derived result are hash-addressed for independent inspection.

Diffusion-rate audit · ICML 2026 Reproduction Challenge · Pinned arXiv source, finite CPU controls, and public GitHub evidence.

Code: **github.com/MachineLearning-Nerd/icml26-repro-L5JTAPUdbQ-diffusion-low-dimensional-distributions**